

# Keyboard-to-Pico wire list

Bread80 CPC464 SMT / J3 19-pin connector / original RP2040 Pico or Pico H

**Keyboard disconnected from CPC.** Use 19 signal wires only. Do not connect a matrix contact to power or ground. Power the Pico through its USB connector.



| J3 pad | Matrix net | Pico GPIO | Pico header pin | Signal role |
|--------|------------|-----------|-----------------|-------------|
| 01     | X9         | GP20      | 26              | Sense input |
| 02     | Y10        | GP11      | 15              | Scan output |
| 03     | X1         | GP12      | 16              | Sense input |
| 04     | X2         | GP13      | 17              | Sense input |
| 05     | X3         | GP14      | 19              | Sense input |
| 06     | X4         | GP15      | 20              | Sense input |
| 07     | X5         | GP16      | 21              | Sense input |
| 08     | X6         | GP17      | 22              | Sense input |
| 09     | X7         | GP18      | 24              | Sense input |
| 10     | X8         | GP19      | 25              | Sense input |
| 11     | Y9         | GP10      | 14              | Scan output |
| 12     | Y8         | GP9       | 12              | Scan output |
| 13     | Y7         | GP8       | 11              | Scan output |
| 14     | Y6         | GP7       | 10              | Scan output |
| 15     | Y5         | GP6       | 09              | Scan output |
| 16     | Y4         | GP5       | 07              | Scan output |
| 17     | Y3         | GP4       | 06              | Scan output |
| 18     | Y2         | GP3       | 05              | Scan output |
| 19     | Y1         | GP2       | 04              | Scan output |

**Header pin is the physical pin number.** For example, J3 pad 1 connects to **GP20 on Pico pin 26**, not to Pico pin 20.

## Identify J3 before wiring

With keys facing you and cursor keys at upper right, J3 is the single vertical row at the keyboard's far left. On the inspected PCB, **pad 1 is square at the Esc end**; pad 19 is at the Caps end. The solder-side view is mirrored.

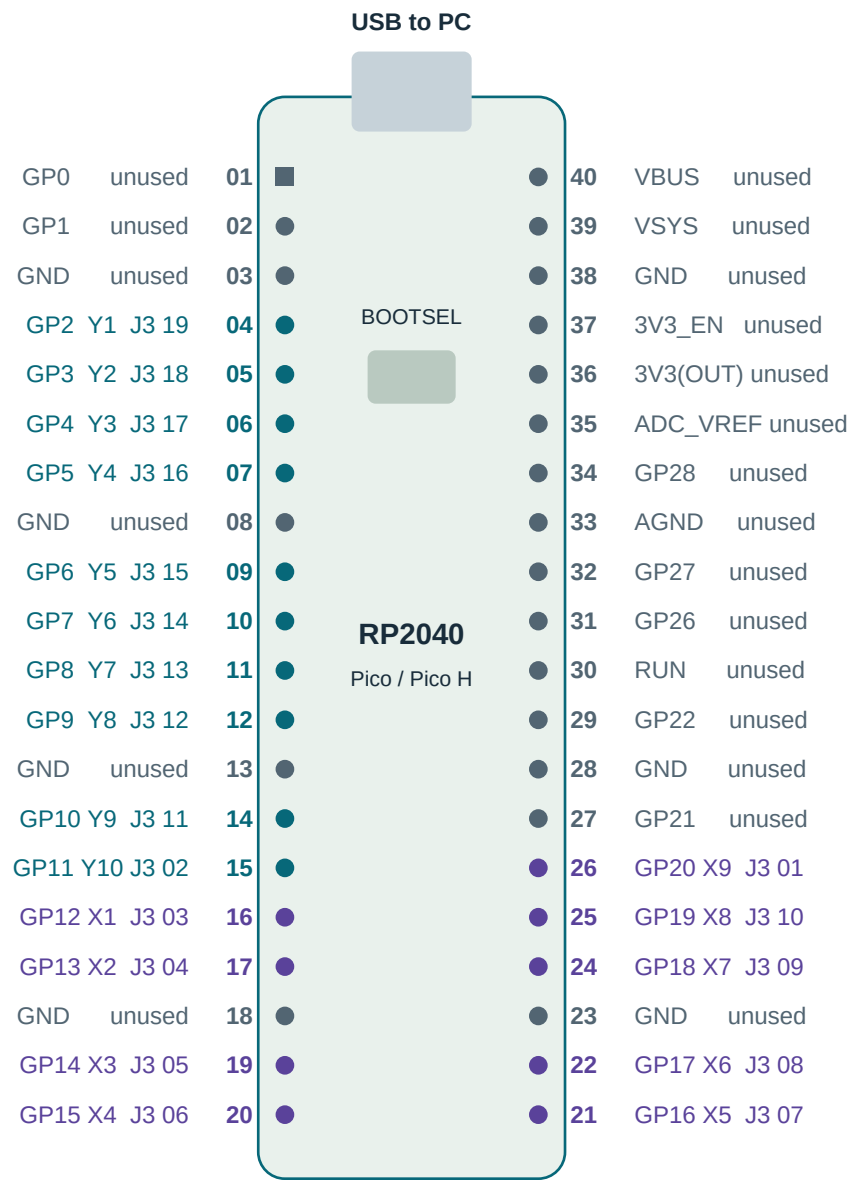
Leave **LK1, LK2, LK3 and LK4-LK6 open** for this prototype. LK3 joins X8/X9; the last three links serve the unused bonus keys. **J1 and J2 have different pin orders.**

J3 nets: Bread80 PCB revision bf4a6052888f. Firmware pin assignments: src/board\_config.h. See page 4 for rgbwalker V1.2 checks and the first hardware test.

# Pico header orientation

Top view: components and BOOTSEL button facing you, USB connector at the top. Not to scale.

Numbers beside the board are **physical header pins**. Labels show GPIO / matrix net / J3 pad. Anything marked **unused** stays unconnected in this keyboard harness.



Y = scan outputs X = sense inputs Grey = unused

## Three details that prevent wiring mistakes

**Count down the left side: 1 to 20.** The right side counts up from 21 at the bottom to 40 at the top. Ground pins interrupt the GPIO sequence.

**No matrix supply pin:** X inputs use the Pico's internal 3.3 V pull-ups. The USB cable powers the Pico; the passive keyboard needs only the 19 signal connections.

**Do not join X8 and X9 in this harness.** X9 is dedicated to DEL. Both Shift switches share Y3/X6 and cannot be distinguished by firmware.

Physical numbering checked against [Raspberry Pi's official Pico pinout PDF](#). The diagram above is an original drawing for this project.

# Assemble and verify

Work with USB unplugged during wiring and meter checks. Connect only the passive keyboard.

- 1. Prepare:** identify the exact board and J3 pad 1. Open LK1-LK6. Fit Pico headers if needed. Use insulated wires or a labeled breakout harness; keep the first cable short.
- 2. Wire:** follow page 1 one contact at a time. Check every cable endpoint and look for unintended adjacent shorts. Do not rely on ribbon-wire colors or a mirrored photograph.
- 3. Test contacts:** disconnect the keyboard from both Pico and CPC; check the pairs below. On a diode board, place the red meter probe on X and black on Y, using diode mode.

| Key held     | Y / X nets | J3 pads | Pico header pins |
|--------------|------------|---------|------------------|
| Cursor Up    | Y1 / X1    | 19 / 3  | 4 / 16           |
| A            | Y9 / X6    | 11 / 8  | 14 / 22          |
| Space        | Y6 / X8    | 14 / 10 | 10 / 25          |
| Either Shift | Y3 / X6    | 17 / 8  | 6 / 22           |
| DEL          | Y10 / X9   | 2 / 1   | 15 / 26          |
| Main Enter   | Y3 / X3    | 17 / 5  | 6 / 19           |
| Keypad Enter | Y1 / X7    | 19 / 9  | 4 / 24           |

A pressed key should conduct; releasing it should open the path. A diode blocks the reverse direction. A plain switch conducts either way. A continuity beeper may not detect the diode path reliably.

**Current path:** internal 3.3 V pull-up → X → closed switch → diode → selected Y.  
Diode cathode faces Y. Only the selected Y is driven low; inactive Y pins float.

## Choose the correct scan option

**Bread80 SMT, all diodes verified:** set the option ON. **rgbwalker, membrane, tactile, or uncertain diodes:** keep it OFF. These commands assume the SDK build is already configured; see README.md for initial setup.

```
cmake -S . -B build -DGOODJOHN_MATRIX_HAS_DIODES=ON
cmake --build build --parallel
```

Use **OFF** instead of ON for a matrix without verified diodes.

## Flash, then test on the PC

Hold BOOTSEL while connecting the Pico, then copy **build/goodjohn.uf2** to RPI-RP2. Use a USB data cable. Check every key and release, modifiers, DEL and both Enter keys. COPY sends Alt; DEL sends Backspace; CLR sends forward Delete. Set the PC's Num Lock for keypad digits.

If one group of keys fails, recheck its X/Y wire and diode orientation. If only chords fail on a board without diodes, ghost suppression may be blocking ambiguity. More than six regular keys produces HID ErrorRollOver. Shifted symbols follow the PC layout; there is no Fn layer yet.

V1.2 digits, all letter rows, Esc and main Enter passed with the completed direct harness; see page 4. Remaining keys, chords, cable settling, USB reconnect and suspend still need tests. No USB remote wakeup or low-power suspend implementation is provided yet.

# Other boards and source notes

The logical CPC key matrix is shared; connector order and jumper routing are not universal.

**rgbwalker V1.2: basic typing passed (2026-10-06).** User confirmed digits, all letter rows, Esc and main Enter with the completed 19-wire direct harness.

V1.2 Gerbers connect CP002\_THT and CP002\_SMD contacts 1-19 in the same order; contact 20 is isolated from the matrix. Count the **square pad as 1 at the Esc end**, toward Shift. Check other revisions separately.

| Expected contact(s) | Expected switch lines       | Pico harness, if confirmed |
|---------------------|-----------------------------|----------------------------|
| 1 and 2             | Isolated DEL switch         | GP20 and GP11              |
| 3 through 10        | X1 through X8               | GP12 through GP19          |
| 11 through 19       | Y9 through Y1               | GP10 down through GP2      |
| 20                  | No matrix connection (V1.2) | Leave isolated             |

**Start directly:** keyboard **3 to Pico physical 16 (GP12)**; keyboard **11 to physical 14 (GP10)**. With USB unplugged, each wire must beep pad-to-pad without a key held. Connect USB: top-row 1 should type 1. Bypassing the earlier ribbon/jumper assembly restored this; its exact fault is unresolved.

Unplug USB, then add **4 to physical 17 (GP13)** for 2; next **5 to physical 19 (GP14)** for Esc. Check each wire before power. Esc prints no character. Complete contacts 1-19 using page 1, leave 20 disconnected, and keep **GOODJOHN\_MATRIX\_HAS\_DIODES=OFF**. Details: docs/wiring.md.

## Bread80 alternatives

**J1 membrane, CPC464 links:** see the separate **docs/pinout-dual-ribbon.pdf** and wiring-dual-ribbon.svg. **J2 modular:** not mapped here. **Tactile:** different connectors, no diodes. **CPC6128 SMT:** design-stage board; not a validated adapter target.

An 18-wire adaptation must handle the shared X8/X9 net explicitly; simply omitting X9 loses DEL.

## Source records

### Bread80 CPC464 SMT KiCad PCB

Revision bf4a6052888fef27ca01a6b0983791d7cb4e8af1. J3 pad nets, square pad 1, connector placement, diode cathodes and switch coordinates were inspected.

### rgbwalker V1.2 fabrication archive

Revision 588e61006bb606d08cc981467d7990c6e3104c42. Copper/plated-hole tracing checks THT/SMD correspondence, unused contact 20, and switch pairs: 1 = 3/11, 2 = 4/11, Esc = 5/11.

### Raspberry Pi official Pico pinout

Pico-R3-A4-Pinout.pdf. Used to check the 40 physical header positions and power/ground labels. Target is the original non-wireless RP2040 Pico/Pico H.

## Files and regeneration

The companion ASCII sheet is **docs/pinout.txt**. The generator reads the firmware GPIO arrays in **src/board\_config.h** and produces both formats from one wiring dataset. These are original vector drawings, not scaled PCB templates.

```
python3 -m pip install -r scripts/requirements-docs.txt
python3 scripts/generate_pinout.py
```